

Q3 A: What is the osculating sphere of a curve? Derive its radius and center in terms of the curvature and torsion of the curve.

Preliminaries: Frenet–Serret Apparatus

Let $\mathbf{r}(s)$ be a smooth curve in $\mathbb{R}^3$ parameterized by arc length s , with $\kappa(s) > 0$ and $\tau(s) \neq 0$. The Frenet–Serret frame $\{\mathbf{T}, \mathbf{N}, \mathbf{B}\}$ satisfies:

$$\mathbf{T}' = \kappa \mathbf{N}, \quad \mathbf{N}' = -\kappa \mathbf{T} + \tau \mathbf{B}, \quad \mathbf{B}' = -\tau \mathbf{N},$$

where primes denote d/ds . Set $\rho = 1/\kappa$ (radius of curvature) and $\sigma = 1/\tau$ (radius of torsion).

Definition

The **osculating sphere** (or *sphere of curvature*) of a curve $\mathbf{r}(s)$ at a point $P = \mathbf{r}(s_0)$ is the unique sphere that has *third-order contact* with the curve at P . Equivalently, it is the limiting position of the sphere passing through four consecutive points on the curve as all four points approach P .

It contains the osculating circle at P as a great circle and lies on the same side of the osculating plane as the centre of curvature.

Derivation of the Centre and Radius

The centre $\mathbf{C}$ and radius R of the osculating sphere at $\mathbf{r}(s)$ are:

$$\mathbf{C} = \mathbf{r} + \rho \mathbf{N} + \rho' \sigma \mathbf{B},$$

$$R = \sqrt{\rho^2 + (\rho' \sigma)^2} = \sqrt{\rho^2 + \left(\frac{\rho'}{\tau}\right)^2}.$$

Proof. Step 1: Parameterize the centre.

Let the centre of the osculating sphere be $\mathbf{C}(s)$, at constant distance R from $\mathbf{r}(s)$. Write the displacement in the Frenet frame:

$$\mathbf{C} - \mathbf{r} = a \mathbf{T} + b \mathbf{N} + c \mathbf{B},$$

so that the sphere condition is:

$$|\mathbf{C} - \mathbf{r}|^2 = a^2 + b^2 + c^2 = R^2. \quad (0)$$

Step 2: First-order contact condition.

Differentiate (0) with respect to s :

$$\frac{d}{ds} |\mathbf{C} - \mathbf{r}|^2 = 0 \implies 2(\mathbf{C} - \mathbf{r}) \cdot \frac{d}{ds} (\mathbf{C} - \mathbf{r}) = 0.$$

Since $\mathbf{C}$ is the centre of the *osculating* sphere (third-order contact), we require $\mathbf{C}$ to be chosen so that this equation and the next two are satisfied. More directly: for contact of order ≥ 1 , the tangent vector $\mathbf{T}$ must be perpendicular to $\mathbf{C} - \mathbf{r}$:

$$(\mathbf{C} - \mathbf{r}) \cdot \mathbf{T} = 0 \implies a = 0. \quad (1)$$

So $\mathbf{C} - \mathbf{r} = b\mathbf{N} + c\mathbf{B}$.

Step 3: Second-order contact condition.

Differentiate $(\mathbf{C} - \mathbf{r}) \cdot \mathbf{T} = 0$ with respect to s :

$$\frac{d}{ds}[(\mathbf{C} - \mathbf{r}) \cdot \mathbf{T}] = 0.$$

Expanding (using $\frac{d(\mathbf{C}-\mathbf{r})}{ds} = -\mathbf{T} + \mathbf{C}'$ and, for an osculating object, the effective differentiation acts via the curve's derivatives):

$$-|\mathbf{T}|^2 + (\mathbf{C} - \mathbf{r}) \cdot \mathbf{T}' = 0 \implies -1 + (\mathbf{C} - \mathbf{r}) \cdot \kappa\mathbf{N} = 0 \implies \kappa b = 1 \implies b = \frac{1}{\kappa} = \rho. \quad (2)$$

Step 4: Third-order contact condition.

We now differentiate the relation $\mathbf{C} - \mathbf{r} = \rho\mathbf{N} + c\mathbf{B}$ with respect to s and impose that the centre $\mathbf{C}$ is fixed (to third order), i.e. $\mathbf{C}' = \mathbf{0}$ in the osculating sense. Differentiating:

$$-\mathbf{T} = \rho'\mathbf{N} + \rho\mathbf{N}' + c'\mathbf{B} + c\mathbf{B}'.$$

Substituting Frenet–Serret formulas:

$$-\mathbf{T} = \rho'\mathbf{N} + \rho(-\kappa\mathbf{T} + \tau\mathbf{B}) + c'\mathbf{B} + c(-\tau\mathbf{N}).$$

Collecting components:

$$\mathbf{T}\text{-component: } -1 = -\rho\kappa = -1. \quad \checkmark$$

$$\mathbf{N}\text{-component: } 0 = \rho' - c\tau \implies c\tau = \rho' \implies c = \frac{\rho'}{\tau} = \rho'\sigma. \quad (3)$$

$$\mathbf{B}\text{-component: } 0 = \rho\tau + c'.$$

The $\mathbf{T}$ -component is automatically satisfied (confirming consistency with Step 3), and equation (3) uniquely determines c .

Step 5: Centre and radius.

Substituting (1), (2), (3):

$$\boxed{\mathbf{C} = \mathbf{r} + \rho\mathbf{N} + \frac{\rho'}{\tau}\mathbf{B}}$$

The radius is:

$$R^2 = |\mathbf{C} - \mathbf{r}|^2 = b^2 + c^2 = \rho^2 + \left(\frac{\rho'}{\tau}\right)^2,$$

$$\boxed{R = \sqrt{\rho^2 + \left(\frac{\rho'}{\tau}\right)^2}}$$

□

Explicit Form in Terms of κ and τ

Since $\rho = 1/\kappa$, we have $\rho' = -\kappa'/\kappa^2$. Therefore:

$$\frac{\rho'}{\tau} = \frac{-\kappa'}{\kappa^2\tau},$$

and the centre and radius become:

$$\mathbf{C} = \mathbf{r} + \frac{1}{\kappa}\mathbf{N} - \frac{\kappa'}{\kappa^2\tau}\mathbf{B},$$

$$R = \sqrt{\frac{1}{\kappa^2} + \frac{\kappa'^2}{\kappa^4\tau^2}}.$$

Geometric Interpretation

1. The centre $\mathbf{C}$ lies in the **normal plane** (perpendicular to $\mathbf{T}$) through $\mathbf{r}$, since the $\mathbf{T}$ -component is zero.
2. The $\mathbf{N}$ -component $\rho = 1/\kappa$ is exactly the radius of curvature, placing $\mathbf{C}$ at distance ρ from the curve in the $\mathbf{N}$ direction — the same as the centre of the osculating circle.
3. The $\mathbf{B}$ -component ρ'/τ measures how far the sphere's centre is displaced *out of* the osculating plane, governed by how quickly the curvature is changing (ρ') relative to the torsion (τ).
4. When $\kappa' = 0$ (constant curvature, e.g. a circle or circular helix), $\rho' = 0$ and the centre reduces to $\mathbf{r} + \rho\mathbf{N}$, which is just the centre of the osculating circle — as expected.
5. When $\tau = 0$ (planar curve), the formula breaks down ($c \rightarrow \infty$), reflecting the fact that for a planar curve the osculating sphere degenerates to the osculating circle (a sphere of infinite radius in the limit).

Summary

$$\text{Centre:} \quad \mathbf{C} = \mathbf{r} + \underbrace{\rho}_{\text{along } \mathbf{N}} \mathbf{N} + \underbrace{\frac{\rho'}{\tau}}_{\text{along } \mathbf{B}} \mathbf{B}$$

$$\text{Radius:} \quad R = \sqrt{\rho^2 + \left(\frac{\rho'}{\tau}\right)^2}$$